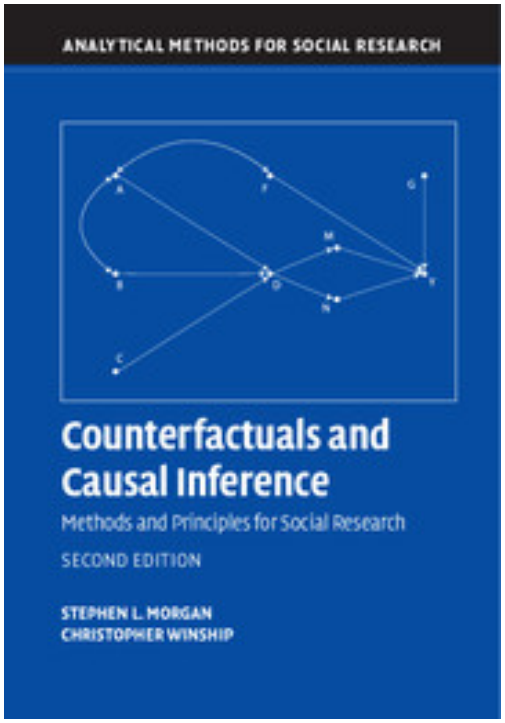
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 4

Outline

1. Conditioning and Directed Graphs
2. The Back-Door Criterion
3. Models of Causal Exposure and Point Identification Based on the Potential Outcome Model
4. Conditioning to Balance and Conditioning to Adjust
5. Appendix to Chapter 4: The Back-Door and Adjustment Criteria, Descendants, and Colliders Under Magnification

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Recall our first look at confounding and conditioning

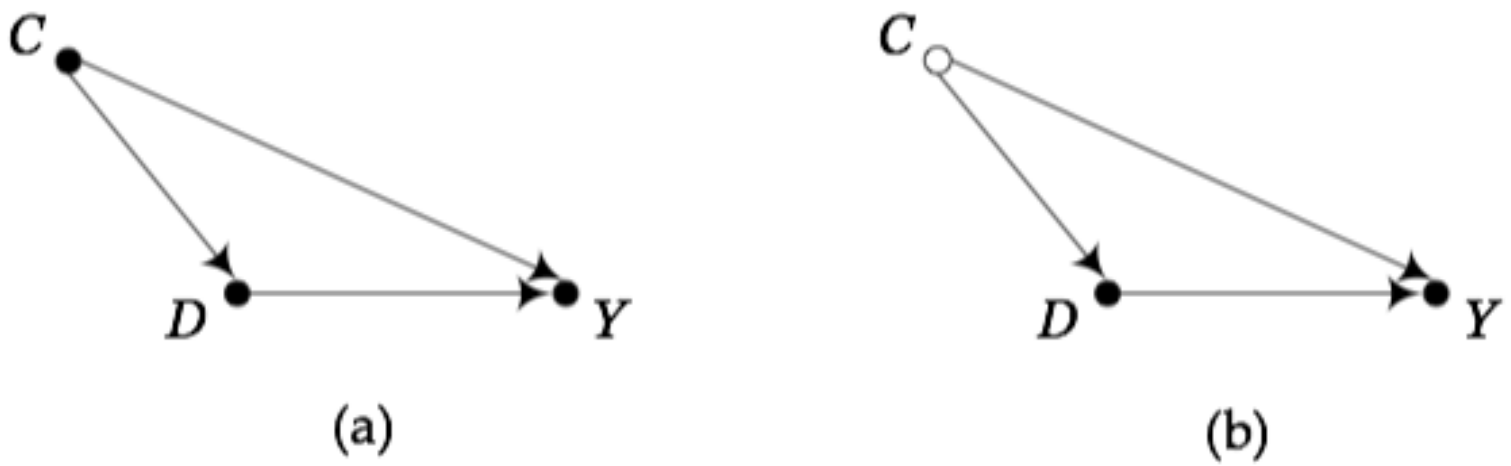

Figure 3.4 Two graphs in which the causal effect of D on Y is confounded by C.

Consider an elaborated version

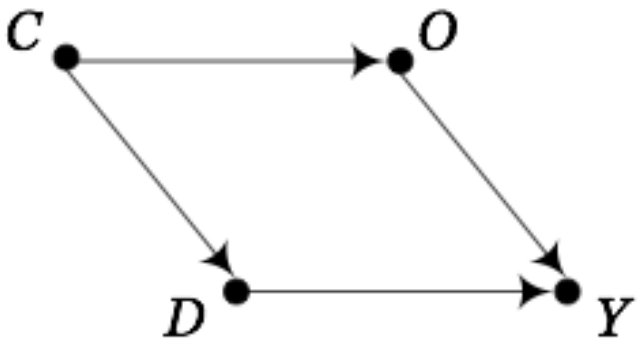

Figure 4.1 A graph in which the causal effect of D on Y is confounded by the back-door path $D \leftarrow C \rightarrow O \rightarrow Y$.

From confounders to back-door paths

- The key goal of a conditioning strategy is not to adjust for any particular confounder but rather to remove the portion of the total association between D and Y that is noncausal.
- For Figure 4.1, the strategy to adjust for C is "balancing the determinants of treatment assignment." The strategy to adjust for O is "adjusting for all other causes of the outcome."
- Pearl characterizes both strategies in a novel way, using the language of back-door paths:
  - A *path* is any sequence of edges pointing in any direction that connects one variable to another.
  - A *back-door path* is a path between any causally ordered sequence of two variables that begins with a directed edge that points to the first variable.

Now consider conditioning and colliders

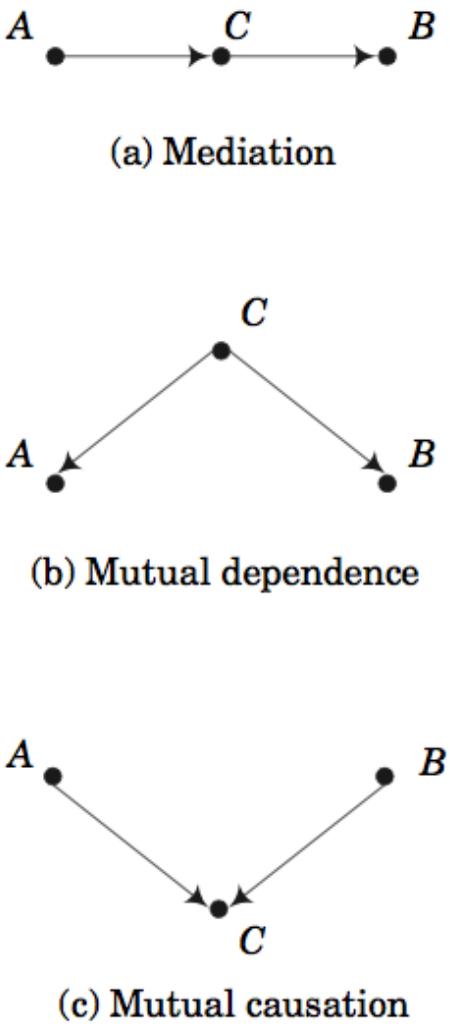

Figure 3.3 Basic patterns of causal relationships for three variables.

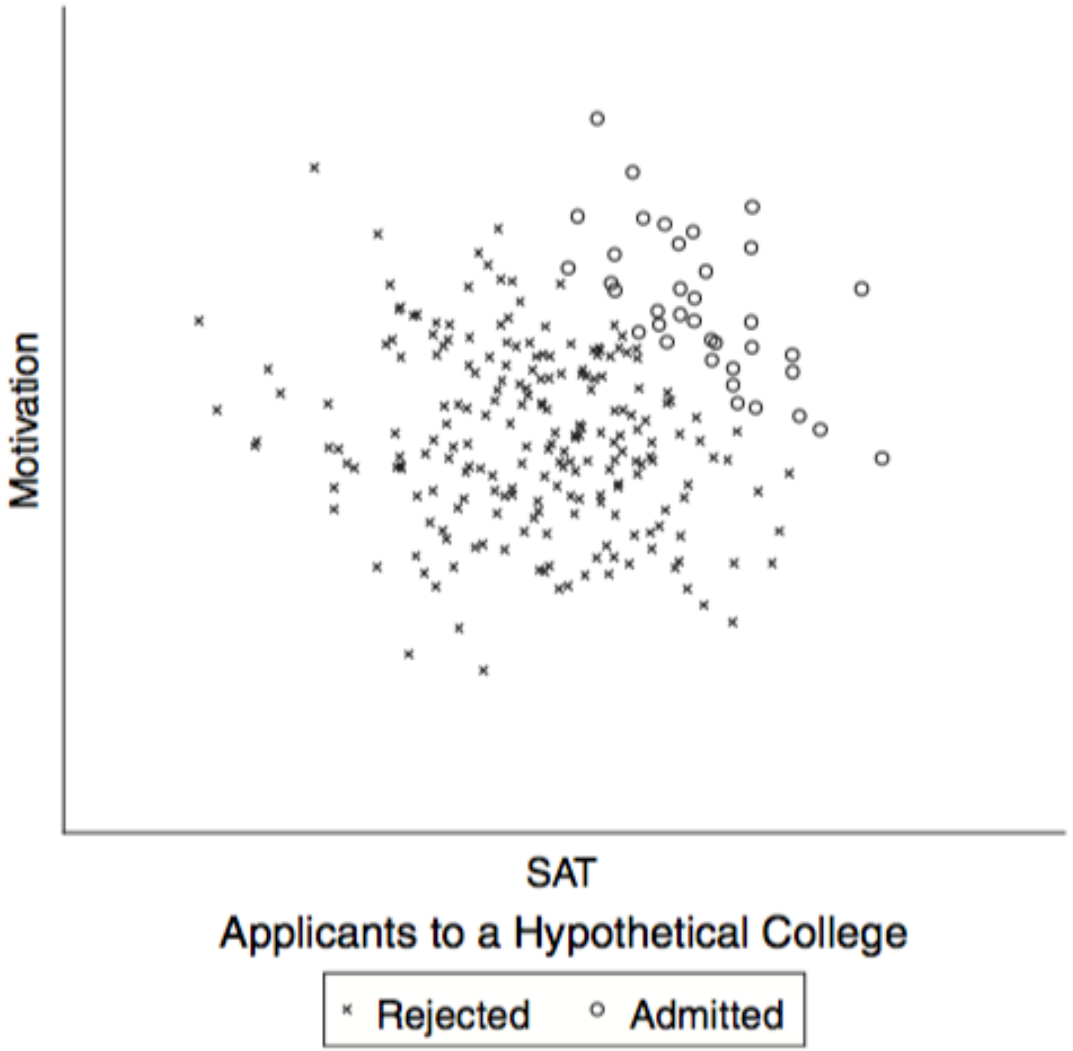

Figure 4.2 Simulation of conditional dependence within values of a collider variable.

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Two steps for back-door conditioning

1. Write down the back-door paths from the causal variable to the outcome variable, determine which ones are unblocked, and then search for a candidate conditioning set of observed variables that will block all unblocked back-door paths.
2. If a candidate conditioning set is found that blocks all back-door paths, inspect the patterns of descent in the graph in order to verify that the variables in the candidate conditioning set do not block or otherwise adjust away any portion of the causal effect of interest.

The back-door criterion

- Condition 1. All back-door paths between the causal variable and the outcome variable are blocked after conditioning on Z, which will always be the case if each back-door path
  - (a) contains a chain of mediation $A \rightarrow C \rightarrow B$, where the middle variable C is in Z, or
  - (b) contains a fork of mutual dependence $A \leftarrow C \rightarrow B$, where the middle variable C is in Z, or
  - (c) contains an inverted fork of mutual causation $A \rightarrow C \leftarrow B$, where the middle variable C and all of C's descendants are *not* in Z;

and

The back-door criterion (cont'd)

- Condition 2. No variables in Z are descendants of the causal variable that lie on (or descend from other variables that lie on) any of the directed paths that begin at the causal variable and reach the outcome variable.

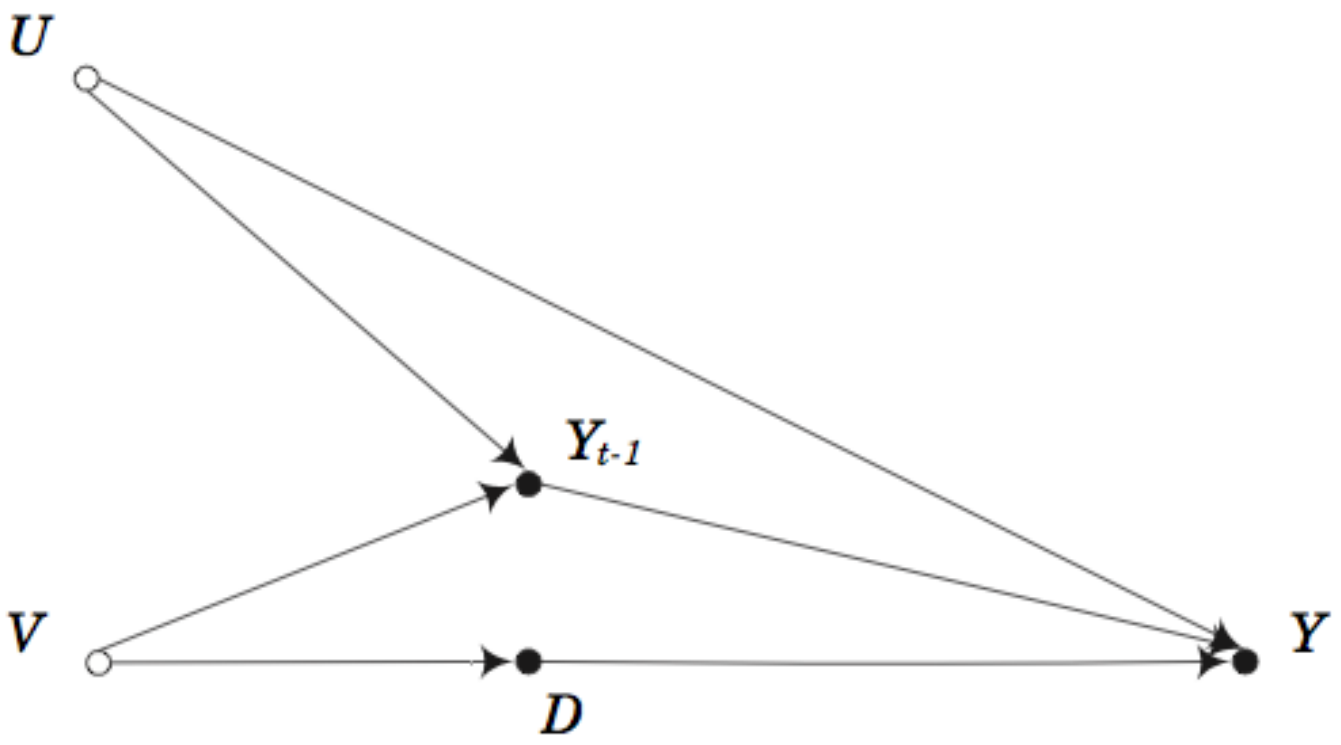

Figure 4.3 A causal diagram in which Y_{t-1} is a collider along a back-door path.

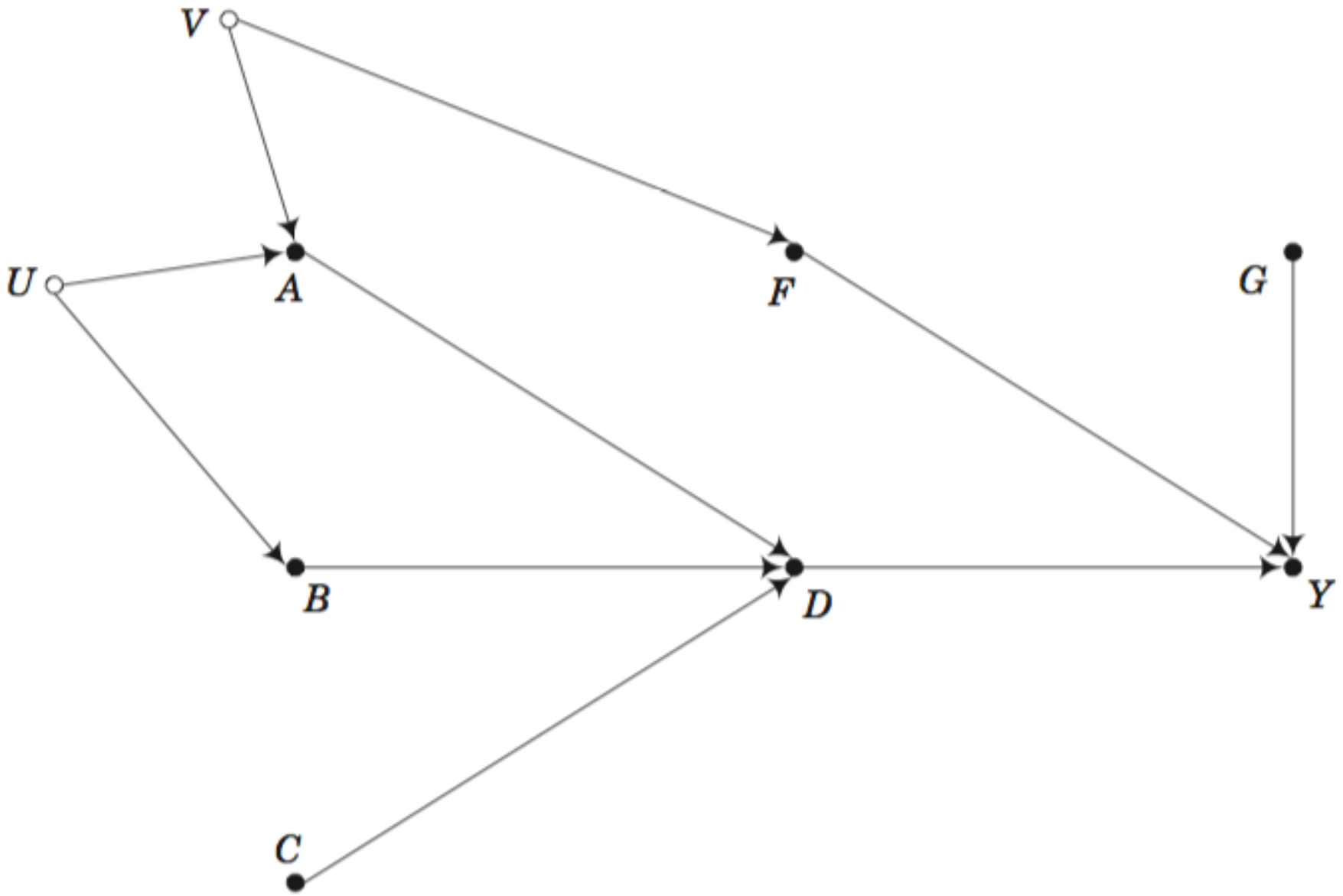

Figure 4.4 A causal diagram in which A is a collider on a back-door path.

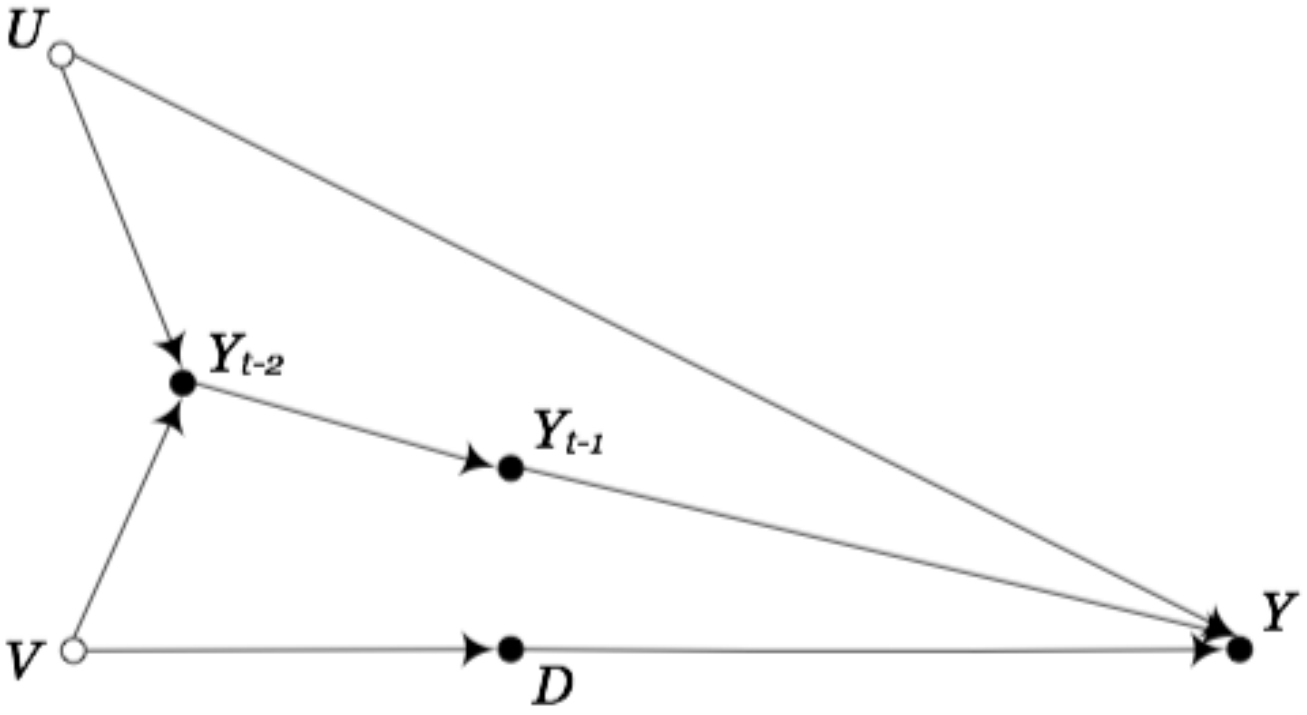

Figure 4.5 A causal diagram in which Y_{t-2} is a collider on a back-door path and Y_{t-1} is its descendent.

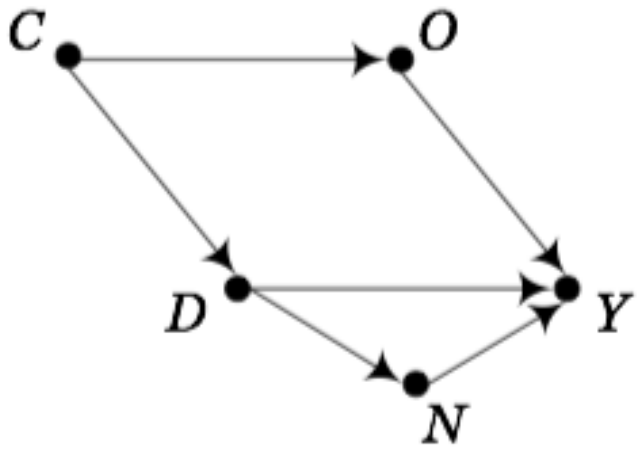

Figure 4.6 A confounded causal effect expressed as an indirect and net a direct effect.

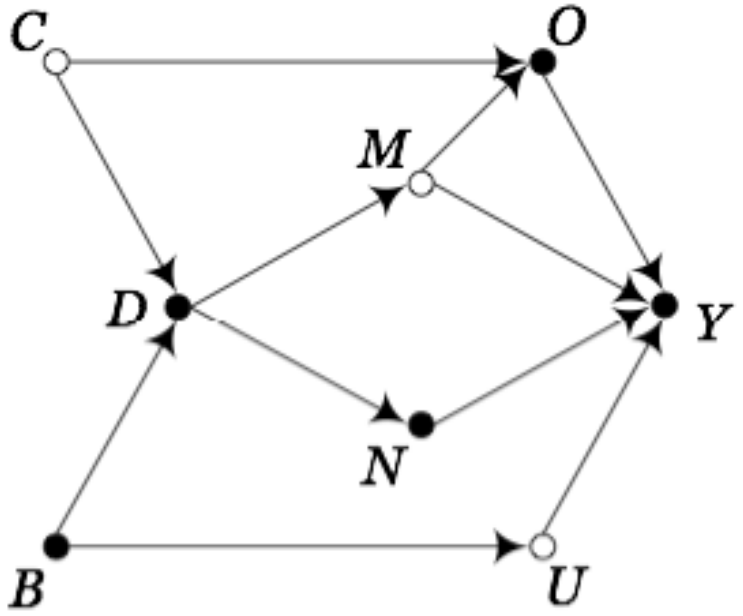

Figure 4.7 A graph where the effect of D on Y is not identified by conditioning on O and B because O is a descendant of D.

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Treatment assignment modeling and randomized experiments

- If treatment assignment is completely randomized by design, then the treatment indicator variable D is completely independent of the potential outcomes Y^0 and Y^1 as well as any function of them, such as the distribution of δ.
- In this case, the treatment assignment mechanism is known because it is set by the researcher who undertakes the randomization.

The probability of treatment assignment can vary

- Consider a design where study subjects are stratified first by gender and then assigned with disproportionate probability to the treatment group if female.
- In this case, the treatment assignment protocol would instead be represented by two conditional probabilities, such as

$$\Pr[D = 1|\text{Gender} = \text{Female}] = .7, \tag{4.1}$$

$$\Pr[D = 1|\text{Gender} = \text{Male}] = .5. \tag{4.2}$$

- These conditional probabilities are typically referred to as "propensity scores," but they are simply conditional probabilities.

Propensity scores in observational research

- A researcher with observational data does not possess a priori knowledge of the exact propensity scores that apply to all individuals.
- Nonetheless, a researcher may know all of the characteristics of individuals that systematically determine their propensity scores.
- Treatment assignment patterns are represented by a general conditional probability distribution,

$$\Pr[D = 1|S], \tag{4.3}$$

where S now denotes all variables that systematically determine all treatment assignment patterns.

Ignorability of treatment assignment

- Complete observation of S allows a researcher to assert that treatment assignment is "ignorable" and then consistently estimate the average treatment effect (ATE)
- Within strata defined by S, the remaining variation in the treatment D is completely random and hence the process that generates this remaining variation is labeled "ignorable."
- The core of the concept of ignorability is the independence assumption that was introduced in Equation (2.6)

$$(Y^0, Y^1) \perp\!\!\!\perp D,$$

where the symbol $\perp\!\!\!\perp$ denotes independence.

Ignorability of treatment assignment (cont'd)

- As defined by Rubin (1978), ignorability of treatment assignment holds when the potential outcomes are independent of the treatment dummy indicator variable, as in this case all variation in D is completely random.
- Ignorability also holds in the weaker case where

$$(Y^0, Y^1) \perp\!\!\!\perp D \mid S \tag{4.4}$$

and

all variables in S are observed.

- The treatment assignment mechanism is ignorable when the potential outcomes (and any function of them, such as δ) are independent of the treatment variable, D, within strata defined by all combinations of values on all variables, S, that systematically determine all treatment assignment patterns.

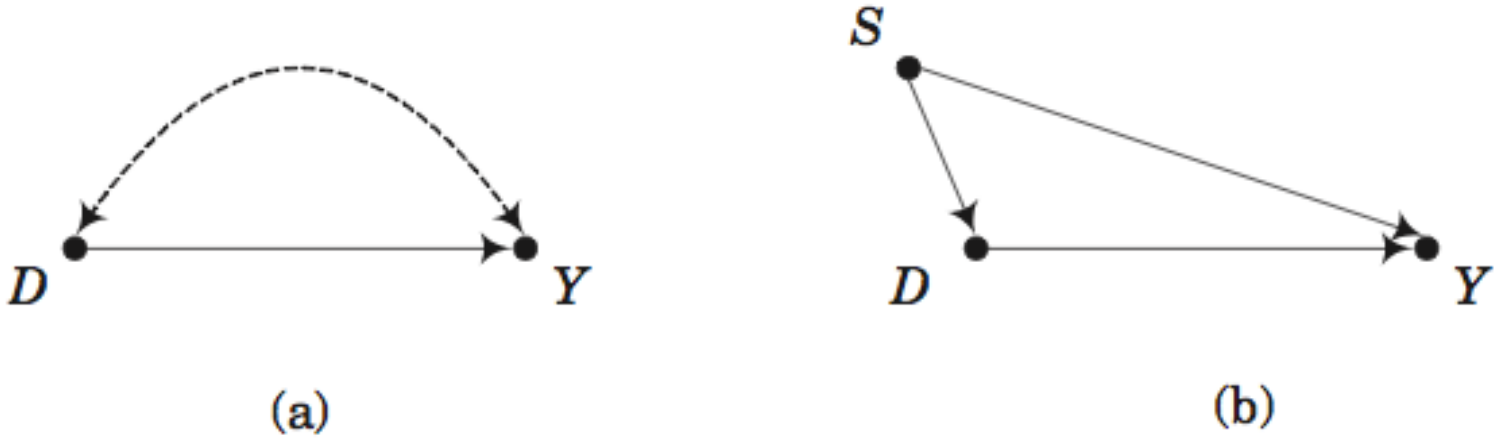


Figure 4.8 Causal diagrams in which treatment assignment is (a) non-ignorable and (b) ignorable.

Treatment selection modeling in econometrics

- Take Equation (2.2),

$$Y = DY^1 + (1 - D)Y^0,$$

and then rearrange and relabel terms as follows:

$$\begin{aligned} Y &= Y^0 + (Y^1 - Y^0)D \\ &= Y^0 + \delta D \\ &= \mu^0 + \delta D + \upsilon^0, \end{aligned} \tag{4.5}$$

where $\mu^0 \equiv E[Y^0]$ and $\upsilon^0 \equiv Y^0 - E[Y^0]$.

- The standard outcome model from the econometrics of treatment evaluation simply reexpresses Equation (4.5) so that potential variability of δ across individuals in the treatment and control groups is relegated to the error term, as in

$$Y = \mu^0 + (\mu^1 - \mu^0)D + \{\upsilon^0 + D(\upsilon^1 - \upsilon^0)\}, \tag{4.6}$$

where $\mu^1 \equiv E[Y^1]$, $\upsilon^1 \equiv Y^1 - E[Y^1]$, and all else is as defined for Equation (4.5)

The treatment selection model

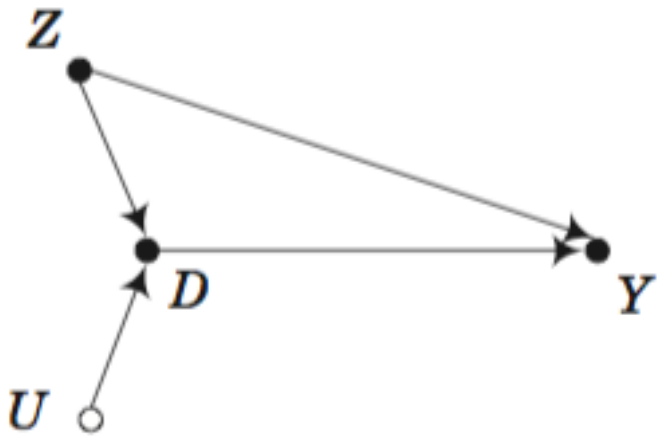
- Treatment selection be modeled by specifying a latent continuous variable $\tilde{D}$ as

$$\tilde{D} = Z\phi + U, \tag{4.7}$$

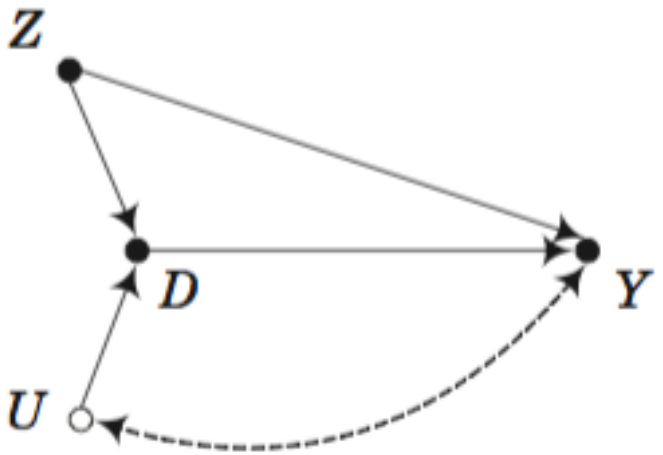
  where Z represents all observed variables that determine treatment selection, ϕ is a coefficient (or a vector of coefficients if Z includes more than one variable), and U represents both systematic unobserved determinants of treatment selection.
- The latent continuous variable $\tilde{D}$ in Equation (4.7) is then related to the treatment selection dummy, D, by

$$\begin{aligned} D &= 1 \text{ if } \tilde{D} \geq 0, \\ D &= 0 \text{ if } \tilde{D} < 0, \end{aligned}$$

  where the threshold 0 is arbitrary because the term U has no inherent metric.



(a) Selection on the observables



(b) Selection on the unobservables

Figure 4.9 Causal diagrams for the terminology from econometric modeling of treatment selection.

Point identification of conditional average treatment effects by conditioning

- Consider a case for which partial ignorability holds, such that $Y^0 \perp\!\!\!\perp D \mid S$ is true but $(Y^0, Y^1) \perp\!\!\!\perp D \mid S$ is not.
- Here, conditioning on S generates a consistent estimate of the ATT even though S does not block the back-door path from D to Y.
- The opposite is, of course, also true. If partial ignorability holds in the other direction, such that $Y^1 \perp\!\!\!\perp D \mid S$holds but $(Y^0, Y^1) \perp\!\!\!\perp D \mid S$ does not, then the ATC can be estimated consistently.

Caveat: Graphs do not clearly reveal the identification possibilities for the ATT and ATC when the ATE is not also identified

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Definition of balance

When treatment status is unassociated with an observed set of variables W, the data are balanced with respect to W. More formally, the data are balanced if

$$\Pr[W|D = 1] = \Pr[W|D = 0], \tag{4.8}$$

which requires that the probability distribution of W be the same within the treatment and control groups.

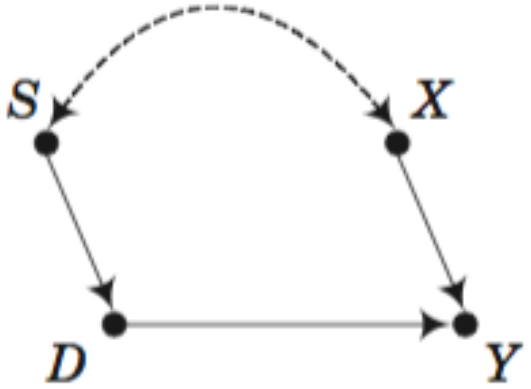

Figure 4.10 A causal diagram in which sufficient conditioning can be performed with respect to S or X.

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The back-door criterion explained with causal graphs viewed under selective magnification

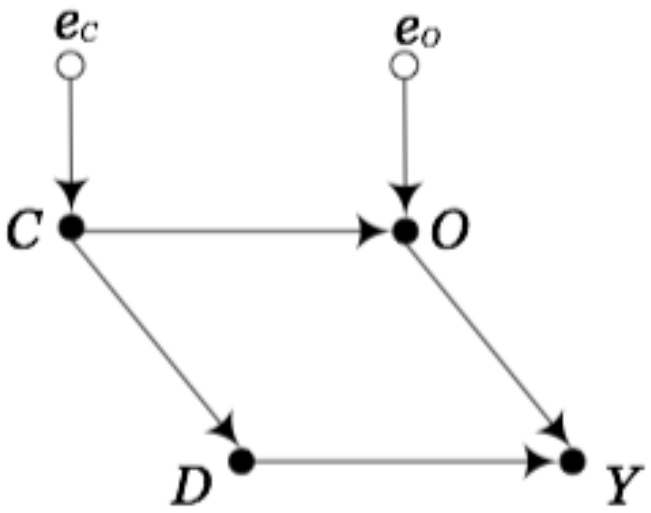

Figure 4.11 A causal graph with a confounded causal effect and where the variables along the back-door path are viewed under magnification.

The important role of Condition 2

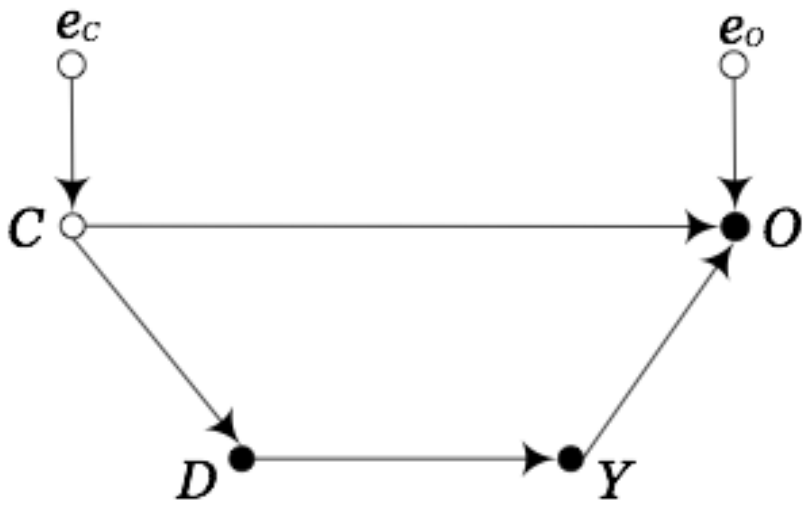

Figure 4.12 A diagram where the causal effect of D on Y is not confounded and where the observed variable O on the back-door path is a descendant of both D and Y.

Why you really should like the back-door criterion: you can ignore complications seen under magnification

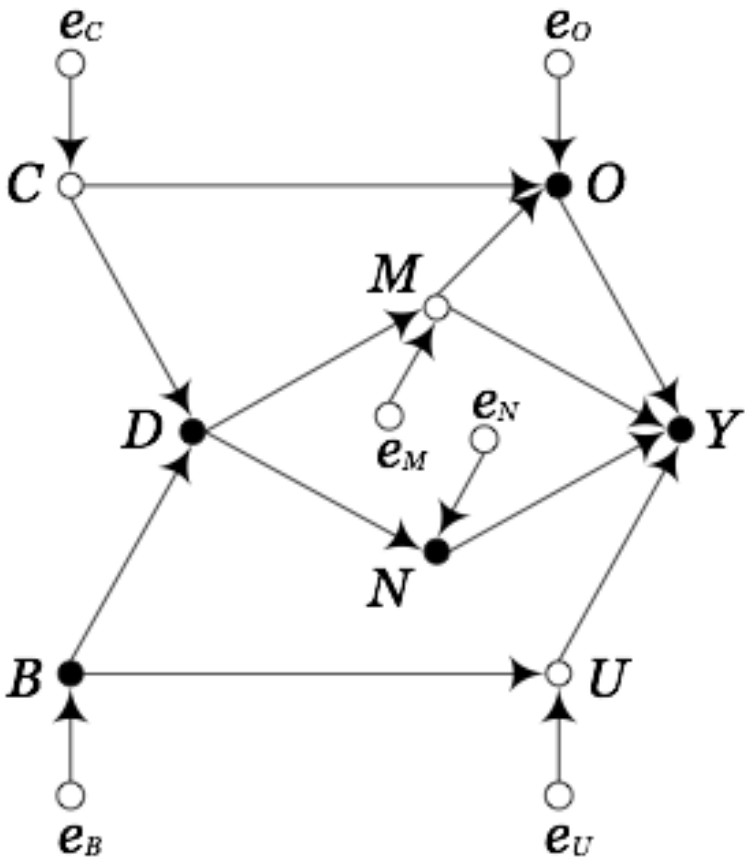

Figure 4.13 A graph where the effect of D on Y is not identified by conditioning on O and B because O is a descendant of D.

The adjustment criterion considered

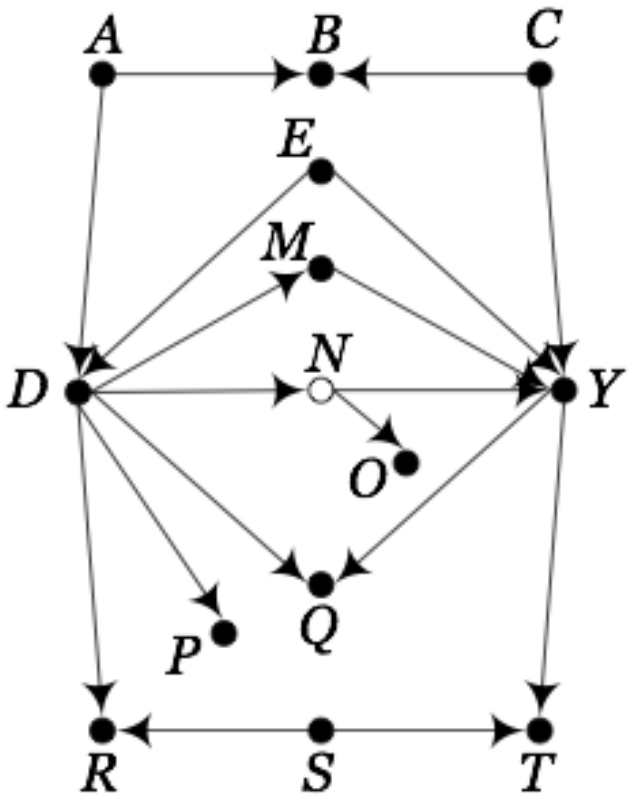

Figure 4.14 A directed graph that reveals the differences between the back-door criterion and the adjustment criterion.